

Course Name: AI/ML/DI and Communication, Batch - II

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Section - A MCQs

Section - B Practical

1) Plan and describe a solution using Azure's Anomaly Detector service to identify unusual patterns in a time-series dataset.

Solution Using Azure Anomaly Detector Service

Azure Anomaly Detector is an Azure AI service used to detect unusual patterns or abnormalities in time-series data such as temperature readings, website traffic, sales, or sensor data.

Step-by-Step Plan

1. Collect Time-Series Data

Prepare the dataset containing:

Timestamp (date/time)

Values (measurements)

Example:

| Time | Sales |

| 10:00 | 120 |

| 11:00 | 125 |

| 12:00 | 300 |

The sudden jump to 300 may indicate an anomaly.

2. Create Azure Anomaly Detector Resource Log in to Azure Portal. Create an Anomaly Detector resource.

Obtain:

Endpoint URL

API Key

These are required to access the service.

3. Send Data to the Service

Use Azure SDK or REST API to send time-series data to the detector.

Two detection methods are available:

Batch Detection → analyzes historical data

Streaming Detection → detects anomalies in real-time

4. Analyze Results

The service returns:

Normal points

Anomalous points

Confidence scores

Expected value ranges

If a data point falls outside the normal pattern, it is flagged as an anomaly.

5. Take Action Based on Detected Anomalies:

Send alerts

Trigger maintenance

Investigate fraud or system issues

Monitor business performance

Conclusion

Azure Anomaly Detector helps automatically identify unusual behavior in time-series datasets, improving monitoring, prediction, and decision-making with minimal manual effort.